THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A cylindrical wire of resistance R is cut into n equal parts which are joined in parallel. The equivalent resistance is 1

(a) R / n^2 **(b)** R / n **(c)** n R **(d)** n^2 R

2. A bar magnet is brought towards a closed coil of wire. The coil 1

(a) becomes a permanent magnet

(b) develops an induced current while the magnet moves

(c) never develops a current

(d) develops a current even if the magnet is at rest

3. A lens produces an erect image 4 times the size of an object placed 7.5 cm from it. Then f is 1

(a) +10 cm **(b)** +7.5 cm **(c)** +5 cm **(d)** -10 cm

4. The size of the particles in the atmosphere responsible for the white colour of clouds is 1

(a) much smaller than the wavelength **(b)** larger than the wavelength of light

(c) zero **(d)** about the size of an atom

5. For the given resistors of 3 ohm, 6 ohm connected in parallel and then in series with 2 ohm and a 6 V cell, the current is 1

(a) 3 A **(b)** 1.5 A **(c)** 1 A **(d)** 0.75 A

6. The average value of a pure alternating current over one complete cycle is 1

(a) half of maximum **(b)** maximum **(c)** zero **(d)** equal to peak

7. Assertion (A): A ray of light passing through the optical centre of a lens does not bend. 1
Reason (R): Thickness of the lens at the centre is negligible and faces are parallel.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): The sky appears blue. Reason (R): Blue light is scattered the most by air molecules. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. Assertion (A): Parallel combination of resistors is used in houses. Reason (R): Each appliance gets the same voltage and works independently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): A current-carrying conductor in a field always experiences a force. Reason (R): The magnetic field acts on the moving charges in the conductor. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Light of speed 1.5×10^8 m/s in a medium has refractive index 1

(a) 1.5 **(b)** 4.5 **(c)** 0.5 **(d)** 2

12. A second identical prism is placed upside down in contact with the first one, in the path of a narrow white beam. The emergent beam is 1

(a) white light **(b)** only red light
(c) only violet light **(d)** a wider spectrum

13. **Assertion (A):** The resistance of a conductor is directly proportional to its length. 1
Reason (R): The resistivity of a material depends on its temperature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A solenoid carrying current behaves like a bar magnet. 1
Reason (R): The magnetic field inside a long solenoid is uniform.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** A ray of light going from air into glass bends towards the normal. 1
Reason (R): The speed of light is greater in glass than in air.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. How does the resistance of a 100 W bulb compare with a 25 W bulb of the same voltage rating? 2

17. Why is a switch always connected to the live wire? 2

18. A convex lens and a concave mirror have the same focal length of 20 cm. Compare the image distance for an object at 40 cm in each case. 2

19. Calculate the power of the lens required by a person whose near point is 50 cm to read at 25 cm. 2

20. A wire of resistance 4 ohm is bent to form a circle. Find its resistance between two diametrically opposite points. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Two wires of the same material and length have radii in the ratio 1 : 2. Find the ratio of their resistances and of the currents when they are connected in parallel across a battery. 3

22. A wire carrying current is bent in the form of a circular loop. How are the field lines inside the loop and outside it different? 3

23. A shopkeeper uses a convex mirror to see the whole shop. A customer is 4 m from the mirror of f = 2 m. Find the image position and size ratio. 3

24. If a person looks at the Sun through a prism in the evening, why does he see colours at the edge of the disc? 3

OR

A student connects a voltmeter in series with the resistor by mistake. What will happen and why? 3

25. Explain the difference between the nature of the magnetic field of a straight wire and a bar magnet. 3

OR

A ray falls normally on a plane mirror. Find the angle of incidence and reflection. Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** In an experiment, a student connects a nichrome wire to a variable battery and notes the readings: when the potential difference is 2 V the current is 0.4 A; at 4 V it is 0.8 A and at 6 V it is 1.2 A. 5

(i) Find the resistance of the wire.

(ii) What is the shape of the V-I graph?

(iii) State Ohm's law.

(iv) The wire is cut into two equal halves and the two halves are joined in parallel. Find the resistance of the combination.

27. A wire of resistance 10 ohm is stretched uniformly to three times its length. Find the new resistance. The stretched wire is then folded into three equal parts and joined in parallel. Find the final resistance. 5

OR

Describe the construction and working of a simple electric motor, giving the role of the split-ring commutator. 5

END OF QUESTION PAPER — **Total: 50 Marks**